

LLM-Based MDS-UPDRS Finger Tapping Severity after Feature Selection

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Abstract—Prior LLM-based approaches to clinical motor assessment pass raw kinematic trajectory CSV data ($\sim 2\,400$ rows per session) into each prompt, consuming $\sim 15\,000$ session tokens per query and requiring large proprietary models. We propose a more token-efficient alternative combining systematic feature selection with few-shot prompting: 149 kinematic features extracted via CoTracker trajectory tracking are evaluated by 28 feature-selection methods spanning filter, embedded, and wrapper families; consensus selection across qualifying methods identifies a 7-feature subset, reducing the total prompt from $\sim 15\,000$ tokens to ~ 400 tokens (zero-shot)—a 97% reduction. Applied to automated MDS-UPDRS Finger Tapping severity scoring ($n=53$ sessions from 16 PD patients), this compact input enables open-source models as small as 32B parameters to surpass GPT-4’s best reported result on the same dataset. The top configuration (Deepseek-R1-32B, few-shot with consensus features) achieves MAE = 0.36, weighted $\kappa = 0.68$, and Adjacent Accuracy = 98%—matching human-expert MAE and approaching inter-rater κ (0.69–0.83). For binary screening (normal vs. abnormal), DS-R1-32B few-shot achieves 100% recall, missing no abnormal session. These findings show that clinically-informed feature selection and pre-extraction can substitute for both raw trajectory input and large proprietary models, enabling cost-efficient clinical assessment.

Index Terms—feature selection, Parkinson’s disease, MDS-UPDRS, token efficiency, clinical NLP

I. INTRODUCTION

Parkinson’s disease (PD) is a progressive neurodegenerative disorder affecting an estimated 11.8 million individuals worldwide as of 2021—a 274% increase in prevalence since 1990 [18]. Bradykinesia, the hallmark motor symptom, is quantified clinically using the MDS-UPDRS Part III Finger Tapping subtest (Item 3.4), which rates speed, amplitude, rhythm, and hesitations on a 0–4 ordinal scale [2]. Because this assessment requires a trained movement disorder specialist, access is limited in resource-constrained settings, motivating automated scoring tools.

Prior machine-learning approaches extract engineered kinematic features (frequency, amplitude, period variability) and feed them into classifiers such as SVM, random forest, or LightGBM [5], [6], [14]. While effective at capturing statistical patterns, these models require hundreds of labeled recordings to train reliably and lack the ability to incorporate domain-specific clinical reasoning [19]. LLMs, trained on vast medical literature, can interpret structured feature data through the lens of clinical guidelines and produce step-by-step rationales

alongside predictions [20]—an advantage particularly salient on the small datasets typical of clinical FT studies.

The closest LLM-based predecessor is Chen [15], who passed *raw 2400-row CSV trajectory data together with a Python feature-extraction script* directly into ChatGPT-4 prompts. This approach required $\sim 15\,000$ session tokens per inference and $\sim 75\,800$ tokens for the four-shot variant (Experiment D), which achieved $\text{ICC}(2,1)=0.45$ —the best result reported on this dataset. Despite high token cost, the reliance on a frontier proprietary model to execute in-context feature engineering limited generalizability.

This paper addresses four gaps left open by prior work. First, most FT deep-learning models require hundreds to thousands of labeled recordings [13], [14], yet clinical datasets are small and expensive to annotate; LLMs operating via in-context learning require no gradient updates, making them uniquely suited to settings with mere tens of samples, where fitting a deep model from scratch is unreliable. Second, no prior study has systematically compared multiple open-weight LLM families—spanning different architectures, reasoning strategies, and parameter scales—on structured ordinal MDS-UPDRS scoring. Third, feature selection for FT has been purely data-driven; no prior work has systematically compared filter, embedded, and wrapper families to derive a consensus subset validated against MDS-UPDRS rubric dimensions. Finally, it remains an open question which kinematic features and which evaluation metrics best reflect clinical judgment for FT severity—leaving both feature design and model evaluation under-specified in the field.

We present a token-efficient pipeline that addresses all four gaps. A 149-feature kinematic pool is extracted from CoTracker sticker trajectories and reduced to a 7-feature consensus subset via systematic comparison of 28 feature-selection methods across filter, embedded, and wrapper families. Pre-extracting features instead of passing raw CSV data reduces the total prompt from $\sim 15\,000$ tokens to ~ 400 tokens (zero-shot)—a **97% reduction**—substantially lowering inference cost and latency for locally-deployed open-weight models. Our key contributions are:

- We benchmark eight open-weight LLMs (7B–72B parameters) across zero-shot and few-shot settings on MDS-UPDRS FT ordinal scoring, establishing the first systematic multi-family comparison on this task.

- We systematically compare 28 feature-selection methods across filter, embedded, and wrapper families; 12 qualifying methods ($\kappa \geq 0.5$, $|S| < 15$) agree on a 7-feature consensus subset that maps directly onto MDS-UPDRS 3.4 rubric dimensions (amplitude, speed, rhythm, hesitations).
- We show that pre-extracted feature prompts achieve a **97%** token reduction versus raw-CSV prompting, enabling open-source 32B-parameter models to surpass Chen’s GPT-4 result ($\kappa=0.68$ vs. ICC=0.45).
- We evaluate binary screening performance (normal vs. abnormal tapping), demonstrating 100% recall—zero missed abnormalities—with DS-R1-32B few-shot, a clinically critical property for a screening tool.

II. RELATED WORK

A. Automated Motor Assessment with Machine Learning

Prior FT assessment systems extract kinematic features (tapping frequency, amplitude, period variability) and train classifiers such as SVM, random forest, or LightGBM on them [5], [6], [14]. These approaches require large labeled datasets and produce scalar predictions without clinical rationales. Feature selection has been purely statistical (ANOVA, Pearson correlation, recursive elimination), with no access to clinical knowledge about what drives FT severity.

B. LLMs for Clinical Scoring

LLMs have demonstrated promising performance across structured clinical reasoning tasks by encoding tabular features as natural language [7]. Zero-shot performance is generally poor on ordinal clinical scales; few-shot prompting with labeled examples substantially improves results by calibrating the model’s internal feature-to-score mapping [3]. Open-weight models—DeepSeek-R1 [24] and Qwen 2.5—now rival proprietary frontier APIs on many reasoning benchmarks, motivating evaluation across multiple model families rather than a single LLM.

The closest predecessor is Chen [15], who evaluated ChatGPT-4 (gpt-4-turbo-2024-04-09) across four experimental conditions (Exp. A–D) on the identical 53-session dataset used here. Zero-shot with raw trajectories (Exp. A) was indistinguishable from chance (ICC(2,1)=0.05); four-shot prompting (Exp. D) reached ICC(2,1)=0.45, accuracy=61%. Each inference required $\sim 15\,000$ session tokens (raw CSV), rising to $\sim 75\,800$ tokens for the four-shot variant. We reproduce Chen’s approach on Gemini Flash to enable a direct metric-space comparison; the reproduced result ($\kappa=0.39$, few-shot) is consistent with Chen’s original ICC and confirms the ceiling imposed by unstructured raw input.

C. Feature Selection for Tabular Data

Feature selection methods fall into three families: *filter* methods rank features by statistical relevance to the target (correlation, mutual information, mRMR); *embedded* methods incorporate selection into model training (regularisation-based and tree-based importance); and *wrapper* methods search the

subset space by directly optimising cross-validated prediction performance [1]. No prior FT study has systematically benchmarked methods across all three families, nor aggregated their outputs via selection-frequency consensus to identify the most robust feature subset.

III. METHOD

A. Dataset and Ethical Approval

The dataset comprises 53 smartphone-recorded finger-tapping sessions from 16 Taiwanese participants with mild-to-moderate Parkinson’s disease (6 women, 10 men; age 69.0 ± 10.14 years; Hoehn & Yahr stage 2.1 ± 0.44) [23]. A *session* is a single 15-second video of one hand performing the task; each participant contributed 1–7 sessions (47% left hand, 53% right hand). Recordings were acquired at Kaohsiung Medical University Hospital (KMUH, Kaohsiung, Taiwan) under IRB approval (No. B-ER-108-362, December 2019) following the apparatus and protocol of Ashyani et al. [21].

Ground-truth MDS-UPDRS scores (0–4) were assigned independently by two movement disorder specialists; the 31 sessions with a one-point disagreement were resolved by rounding the arithmetic mean, yielding a *consensus score* per session (inter-rater weighted $\kappa=0.69$ –0.83). The consensus distribution is: score 0 – 12 sessions (22.6%), score 1 – 17 (32.1%), score 2 – 22 (41.5%), score 3 – 1 (1.9%), score 4 – 1 (1.9%). The dataset is split 70/30 for feature-selection training and held-out evaluation.

B. Trajectory Extraction

Sticker trajectories were extracted from smartphone video (Samsung Galaxy S7 edge, 240 fps, 1280×720 px) using CoTracker [22], a transformer-based point-tracking model. A red sticker on the thumb and a blue sticker on the index finger provided high-contrast markers against a green filming-box background. For each session, the most unobstructed of three synchronized views was selected; two sessions were excluded due to persistent occlusion, yielding the 53 usable recordings.

The Euclidean inter-sticker distance at each frame t is:

$$D(t) = \sqrt{(X_{\text{thumb}} - X_{\text{index}})^2 + (Y_{\text{thumb}} - Y_{\text{index}})^2}, \quad (1)$$

smoothed with a Savitzky–Golay filter (window 11 frames, order 3) and min–max normalised to $[0, 1]$, yielding $\hat{D}(t)$. Individual taps are local maxima of $\hat{D}(t)$ detected with minimum height 0.20, minimum inter-peak interval 0.15 s, and prominence threshold 0.15, providing tap count N , inter-tap periods $\{T_i\}$, and peak amplitudes $\{A_i\}$.

C. Problem Formulation

We organise the data as a *regressor matrix* $\mathbf{X} \in \mathbb{R}^{n \times p}$, where row $\mathbf{x}_i^\top \in \mathbb{R}^p$ contains the p kinematic features extracted from session i , and $\mathbf{y} = [y_1, \dots, y_n]^\top \in \{0, 1, 2, 3, 4\}^n$ is the vector of consensus MDS-UPDRS scores (the *regressand*). With $n = 53$ sessions and $p = 149$ candidate features, the regression problem is to learn a mapping $f : \mathbb{R}^p \rightarrow \{0, 1, 2, 3, 4\}$ that minimises prediction error on unseen sessions.

Feature selection identifies a sparse index set $\mathcal{S} \subseteq \{1, \dots, p\}$ such that $\mathbf{X}_{\mathcal{S}}$ (the sub-matrix formed by the columns indexed by $\mathcal{S}$) is sufficient for accurate prediction while $|\mathcal{S}| \ll p$.

D. Feature Engineering

A total of 149 kinematic features were extracted per session across seven clinically motivated groups (Table I). Throughout this section the following conventions apply. Subscripts: *th*=thumb, *pk*=tap peak, *v_b/v_a*=valley immediately before/after a peak, *op/cl*=opening/closing phase. $D(t)$ is the raw inter-sticker distance (Eq. 1); $\hat{D}(t)$ its min-max normalisation. *6 stats* denotes the set {median, IQR, mean, min, max, std} computed over the per-tap sequence of the named quantity; *9 stats* (Group B frequency) extends this with the linear-fit slope, R^2 , and model degree; *3 stats* (Group G) denotes {median, IQR, mean}. Amplitude decrement symbols: $R_A = A_{\text{late}}/A_{\text{early}}$ (ratio of last-3 to first-3 tap amplitudes); $k_{\text{collapse}} = \text{first tap index where } A_k < 0.5 A_{\text{early}}$; $D_A^{\%} = 100(1 - R_A)$. Freezing symbols: $S_{\text{halt}} = \text{total cumulative halt duration (s)}$; $F_{\text{halt}} = S_{\text{halt}}/T_{\text{total}}$.

E. Feature Selection

To identify which subset of the 149-feature pool best supports MDS-UPDRS 3.4 scoring without committing to a single selection method, we compared 28 feature-selection methods spanning three families: *filter* methods (Pearson, Spearman, Kendall, Mutual Information, F-regression, mRMR [16], Distance Correlation, HSIC, Fisher Score, ReliefF, FCBF, Variance Threshold), *wrapper* methods (RFECV with RF, SVR, and Lasso; SFS Forward; SBS Backward), and *embedded* methods (Standard LASSO, ElasticNet, Adaptive LASSO, LARS, RF Importance, Extra Trees, Gradient Boosting, XGBoost, Permutation Importance, Stability Selection [17], Boruta).

a) Objective functions.: Each FS family optimises a different objective expressed in terms of $\mathbf{X}$ and $\mathbf{y}$.

Filter methods assign each feature j an independent relevance score and retain the top- k . Correlation-based filters use

$$s_j^{\text{corr}} = |\rho(\mathbf{x}_j, \mathbf{y})| \quad (\text{Pearson / Spearman / Kendall}), \quad (2)$$

and Mutual Information uses $s_j^{\text{MI}} = I(\mathbf{x}_j; \mathbf{y})$. mRMR [16] jointly maximises relevance and minimises feature redundancy:

$$\max_{\mathcal{S}} \left[\frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} I(\mathbf{x}_j; \mathbf{y}) - \frac{1}{|\mathcal{S}|^2} \sum_{j, k \in \mathcal{S}} I(\mathbf{x}_j; \mathbf{x}_k) \right]. \quad (3)$$

Embedded methods incorporate selection into model training. LASSO minimises

$$\mathcal{L}_{\text{LASSO}}(\beta) = \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1, \quad (4)$$

where $\lambda > 0$ controls sparsity; non-zero entries of $\hat{\beta}$ identify selected features. ElasticNet adds an ℓ_2 term to (4):

$$\mathcal{L}_{\text{EN}}(\beta) = \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right). \quad (5)$$

Tree-based methods (RF, Extra Trees, Gradient Boosting) rank feature j by its mean impurity reduction across all splits t in the ensemble:

$$s_j^{\text{tree}} = \frac{1}{T} \sum_{t: j_t=j} \frac{|\mathcal{D}_t|}{n} \Delta I(\mathcal{D}_t), \quad (6)$$

where $\Delta I(\mathcal{D}_t) = I(\mathcal{D}_t) - \frac{|\mathcal{D}_L|}{|\mathcal{D}_t|} I(\mathcal{D}_L) - \frac{|\mathcal{D}_R|}{|\mathcal{D}_t|} I(\mathcal{D}_R)$ and $I(\cdot)$ is the node variance for regression. Stability Selection [17] runs LASSO on B bootstrap sub-samples and ranks features by selection frequency $\hat{\Pi}_j = B^{-1} \sum_{b=1}^B \mathbf{1}[\hat{\beta}_j^{(b)} \neq 0]$; only features with $\hat{\Pi}_j \geq \pi_{\text{thr}}$ are retained. Permutation Importance scores feature j by the drop in model performance when its values are randomly shuffled:

$$s_j^{\text{perm}} = \text{score}(\mathbf{X}) - \mathbb{E}_{\sigma} [\text{score}(\mathbf{X}_{\sigma_j})], \quad (7)$$

where $\mathbf{X}_{\sigma_j}$ denotes the data matrix with column j permuted.

Wrapper methods search the subset space by optimising cross-validated prediction error. RFECV backward-eliminates the least-ranked feature at each step:

$$\mathcal{S}^{(k-1)} = \mathcal{S}^{(k)} \setminus \left\{ \arg \min_{j \in \mathcal{S}^{(k)}} r_j \right\}, \quad (8)$$

where r_j is the base-estimator ranking (feature importance for RF/SVR, $|\hat{\beta}_j|$ for LASSO-RFECV), stopping when CV error ceases to improve. SFS greedily adds the feature that most reduces CV error:

$$\mathcal{S}^{(k+1)} = \mathcal{S}^{(k)} \cup \left\{ \arg \min_{j \notin \mathcal{S}^{(k)}} \text{CV-error}(\mathcal{S}^{(k)} \cup \{j\}) \right\}. \quad (9)$$

In all three families the output is a binary selection mask $\mathcal{S}$; the final severity prediction is produced by an MLP retrained on $\mathbf{X}_{\mathcal{S}}$.

b) Evaluation protocol.: Each method's selected subset $\mathcal{S}$ is used to train a fixed MLP regressor (single hidden layer, 8 units, ReLU activation, Adam solver, $\alpha = 1.0$, $\text{max_iter}=600$) on $\mathbf{X}_{\mathcal{S}}$ under leave-one-out cross-validation ($n = 53$). The LOOCV predictions $\hat{\mathbf{y}} = [\hat{y}_1, \dots, \hat{y}_n]^T$ are rounded and clipped to $\{0, 1, 2, 3, 4\}$. Using a fixed predictor isolates the contribution of feature selection from model choice. Each method is ranked by the quadratic-weighted Cohen's κ between $\hat{\mathbf{y}}$ and $\mathbf{y}$:

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{ij} p_{ij}}{\sum_{i,j} w_{ij} p_{i+} p_{+j}}, \quad w_{ij} = (i - j)^2, \quad (10)$$

where p_{ij} is the fraction of observations in confusion-matrix cell (i, j) , and p_{i+} , p_{+j} are the corresponding row and column marginals.

c) Internal estimator vs. MLP prediction.: Both wrapper and embedded methods possess an *internal estimator* that is intrinsically coupled to their selection mechanism. For wrapper methods, this is the base estimator used to evaluate each candidate subset during recursive elimination or sequential search (Random Forest regressor for RFECV-RF, linear SVR for RFECV-SVR, Ridge for SFS/SBS, Lasso for RFECV-Lasso, and Random Forest regressor for Boruta). For embedded methods, the internal estimator is the model whose training

TABLE I
FEATURE GROUPS: DOMAIN AND KEY FORMULAS (149 TOTAL)

Grp	Domain	Key Formula / Definition	N
A	Kinematics	6 stats $\times \{ \Delta X_{th} , \Delta Y_{th} , \ \Delta \mathbf{p}_{th}\ \}$ for thumb & index (36); $ d^2 D/dt^2 , d^2 \dot{D}/dt^2 $ (12)	48
B	Speed	$f_k=1/ITI_k$ (9 stats); $V_{op,k}=\max_t \dot{D}$, $V_{cl,k}=\min_t \dot{D}$ (12); $R_V=\text{med}(V[-3:])/ \text{med}(V[:3])$ (2)	35
C	Amplitude	$A_k=D_{pk_k}-\frac{D_{v_b}+D_{v_a}}{2}$; $\tilde{A}_k=A_k/(D_{\max}-D_{\min})$; R_A , k_{collapse} , $D_A^{\%}$	31
D	Rhythm	$ITI_k=t_{pk_{k+1}}-t_{pk_k}$; $I_{ITI}=\frac{1}{N-2}\sum_k \Delta ITI_k $; aperiodicity, periodEntropy	14
E	Freezing	Interruption: $ITI>1.5\times \text{med}$; Freeze: $ITI>3\times \text{med}$; S_{halt} , F_{halt}	7
F	Task Compl.	N_{valid} , C_{10} , T_{10} , $F_{\text{inc}}=\max(0, 1-N_{\text{valid}}/10)$	5
G	Phase Asym.	$R_{OC,k}=T_{op,k}/T_{cl,k}$; 3 stats each of T_{op} , T_{cl} , R_{OC} ($R_{OC}<1$: prolonged closing)	9
Total			149

objective drives sparsity or importance ranking (LassoCV for LASSO / Adaptive LASSO / Stability Selection, LassoLarsCV for LARS, ElasticNetCV for ElasticNet, and the corresponding tree-based regressors for RF Importance, Extra Trees, Gradient Boosting, XGBoost, and Permutation Importance).

A special case is RFECV with Random Forest, which was originally implemented as a classifier trained on rounded integer labels $\tilde{y}_i = \text{round}(y_i) \in \{0, 1, 2, 3, 4\}$; for the purpose of this comparison its internal predictions are converted to a continuous expected score via $\hat{y}_i = \sum_{c=0}^4 c \cdot \hat{P}(Y=c | \mathbf{x}_i)$, preserving the original classification design while enabling metric-consistent evaluation.

For each method, we run leave-one-out cross-validation *twice* on the same selected feature subset $\mathbf{X}_S$: once with the method’s own internal estimator to obtain $\hat{\mathbf{y}}_{\text{int}}$, and once with the fixed MLP to obtain $\hat{\mathbf{y}}_{\text{MLP}}$. Both prediction vectors are rounded to $\{0, 1, 2, 3, 4\}$ before metric computation. The per-session integer residuals $r_i^{\text{int}} = \text{round}(\hat{y}_i^{\text{int}}) - \text{round}(y_i)$ and $r_i^{\text{MLP}} = \text{round}(\hat{y}_i^{\text{MLP}}) - \text{round}(y_i)$ are then compared jointly, with sessions sorted by consensus score y_i . A method whose internal estimator performs poorly while the MLP performs well on the same subset would signal that the selected features are informative yet the internal model is a poor judge of them, raising concern that the selection may be driven by estimator-specific noise rather than clinical signal. Conversely, consistent agreement between the two predictions supports the reliability of the MLP-based ranking used throughout this study.

d) Selection-frequency consensus.: Rather than committing to a single method’s output, we retained only features judged informative across methods. The qualifying method set is

$$\mathcal{Q} = \{m \in \{1, \dots, 28\} : \kappa_m \geq 0.5, |S_m| < 15\}, \quad (11)$$

and the selection frequency of feature j across qualifying methods is

$$\hat{\Pi}_j = \frac{1}{|\mathcal{Q}|} \sum_{m \in \mathcal{Q}} \mathbf{1}[j \in S_m]. \quad (12)$$

Features are ranked in descending order of $\hat{\Pi}_j$.

e) Optimal input size.: The number of consensus features given to the LLM, N , was treated as a hyperparameter. Candidate values of N were swept exhaustively; for each N , the top- N features (by $\hat{\Pi}_j$) were embedded in the LLM prompt and DeepSeek-R1-32B was queried under few-shot conditions on all 53 sessions. The optimal size is

$$N^* = \arg \max_N \kappa_{\text{LLM}}(\text{top-}N(\hat{\Pi})). \quad (13)$$

This empirical sweep separates the question of *which* features are informative (resolved by the FS comparison) from *how many* an LLM can productively integrate, which depends on the model’s in-context-reasoning capacity.

The consensus-selected features used in all LLM experiments are listed in Table II; they align directly with the MDS-UPDRS 3.4 rubric dimensions of amplitude, speed, rhythm, and hesitations.

TABLE II
CONSENSUS-SELECTED FEATURES (TOP- $N^*=7$ BY $\hat{\Pi}_j$, 12 QUALIFYING METHODS)

Feature	Grp	Clinical Meaning
A_late	C	Median amplitude of last 3 taps
V_open_norm_std	B	Std of normalised opening velocity
period_quartile_range	D	IQR of inter-tap intervals
finger_mvmt_thumb_dist_stddev	A	Std of thumb distance displacement
amplitude_norm_median	C	Normalised median tap amplitude
finger_mvmt_thumb_x_stddev	A	Std of thumb horizontal displacement
finger_mvmt_thumb_x_max	A	Max thumb horizontal displacement

F. LLM Scoring Pipeline

a) Prompt design.: Each prompt contains four elements: (1) task description explaining the data format and collection protocol; (2) the full MDS-UPDRS 3.4 scoring rubric; (3) the consensus-selected kinematic features for the target session; and (4) a score-determination instruction asking for a detailed rationale followed by a single integer (0–4) on the last line. **Zero-shot**: the four elements above, no examples.

Few-shot: extends zero-shot by prepending four labeled reference sessions—one per score level 0–3—each containing both

the raw tap data (periods and amplitudes) and the selected features. Reference sessions were drawn from concordant sessions (both neurologists agree), prioritizing those most frequently mispredicted in a prior pilot study.

b) Inference procedure.: To account for LLM stochastic variation, each session is evaluated 10 times independently under fresh conversation contexts with the default sampling temperature. The session score is the median of the 10 outputs, rounded to the nearest integer in $\{0, 1, 2, 3, 4\}$.

IV. EXPERIMENTS

A. Models

We evaluate the open-source models listed in Table III in zero-shot and few-shot settings using the consensus-selected feature subset. GPT-4-class performance is referenced from published benchmarks on equivalent structured clinical scoring tasks for contextual comparison.

TABLE III
MODELS EVALUATED

Model	Params	Type
Qwen 2.5-7B	7B	Open-source
Qwen 2.5-32B	32B	Open-source
Qwen 2.5-72B	72B	Open-source
Qwen3-8B	8B	Open-source
Qwen3-30B	30B	Open-source
Deepseek-R1-32B	32B	Open-source
Deepseek-R1-70B	70B	Open-source
Mistral-7B	7B	Open-source

B. Evaluation Metrics

Following MDS-UPDRS assessment conventions and prior ordinal scoring literature, we report:

- **MAE ($\downarrow$)**: Mean absolute error between predicted and true score.
- **Weighted κ ($\uparrow$)**: Cohen’s weighted kappa with quadratic weights, measuring ordinal agreement [11].
- **Adjacent Accuracy (ACC) ($\uparrow$)**: Proportion of predictions within ± 1 of the true score.

C. Baselines

- **Human expert (Dr. Tan)**: MAE=0.40, κ =0.69
- **Human expert (Dr. Chien)**: MAE=0.36, κ =0.83

V. RESULTS

A. Feature Selection Quality

Figure 1 plots weighted κ against the number of selected features for all 28 FS methods. Methods in the bottom-right region (dashed lines: $\kappa \geq 0.5$, $|\mathcal{S}| < 15$) pass the Stage 5 quality filter; 15 of 28 methods qualify.

TABLE IV
TOP-10 CONSENSUS FEATURES ACROSS QUALIFYING FS METHODS
($\kappa \geq 0.5$, $|\mathcal{S}| < 15$; 12 OF 28 METHODS PASS).

Feature	Count	Freq. (%)
A_late	11	92%
V_open_norm_std	11	92%
period_quartile_range	7	58%
finger_mvmnt_thumb_dist_stddev	6	50%
amplitude_norm_median	6	50%
finger_mvmnt_thumb_x_stddev	5	42%
finger_mvmnt_thumb_x_max	5	42%
amplitude_norm_mean	4	33%
amplitude_min	4	33%
V_close_norm_std	3	25%

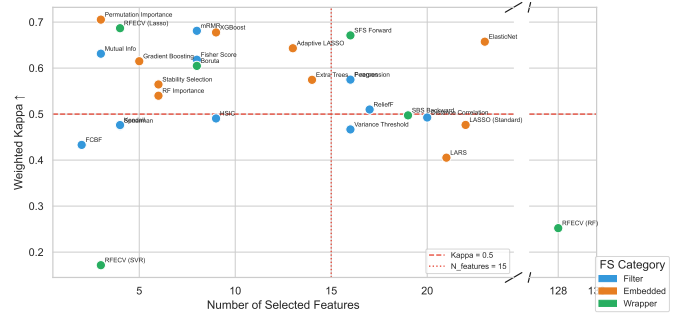


Fig. 1. Number of selected features vs. weighted κ for all 28 FS methods evaluated under LOO-CV with the fixed MLP predictor. Dashed lines mark the Stage 5 inclusion threshold ($\kappa \geq 0.5$, $|\mathcal{S}| < 15$); 15 methods pass. Embedded and wrapper methods cluster in the high- κ region, while most pure-filter methods fall below the threshold.

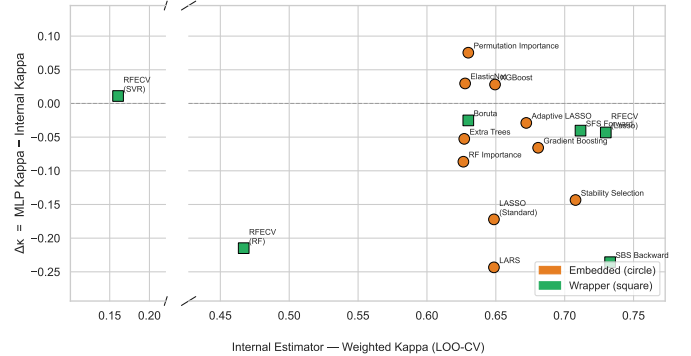


Fig. 2. $\Delta\kappa = \kappa_{\text{MLP}} - \kappa_{\text{int}}$ versus internal-estimator κ for all 16 embedded (circles) and wrapper (squares) FS methods evaluated under LOO-CV on their own selected features. The x-axis is broken to accommodate the RFECV-SVR outlier ($\kappa_{\text{int}} = 0.15$). No method occupies the upper-left quadrant (low internal κ , high $\Delta\kappa$), confirming that MLP-based rankings are not inflated by noise-driven selection.

B. Internal Estimator vs. MLP Prediction on Selected Features

Figure 2 plots $\Delta\kappa = \kappa_{\text{MLP}} - \kappa_{\text{int}}$ against the internal-estimator κ for all 16 embedded and wrapper methods, where each estimator is evaluated by LOO-CV on its own selected feature subset.

Across all 16 methods, 14 show $\Delta\kappa = \kappa_{\text{MLP}} - \kappa_{\text{int}} \leq 0$, meaning the internal estimator performs at least as well as the MLP on its own selected subset. The only exceptions are Permutation Importance ($\Delta\kappa = +0.06$) and Gradient Boosting ($\Delta\kappa = +0.01$), where the MLP modestly outperforms the internal model. Crucially, no method exhibits the pathological pattern of concern—poor internal performance paired with strong MLP performance—that would indicate noise-driven feature selection. The most pronounced gaps occur for linear models (LARS, $\Delta\kappa = -0.40$; LASSO, $\Delta\kappa = -0.26$; SFS Forward, $\Delta\kappa = -0.20$): these methods select features that are highly informative for their own linear assumptions but less exploitable by a non-linear MLP. RFECV-SVR selects only three features and performs poorly under both estimators ($\kappa_{\text{int}} = 0.15$, $\kappa_{\text{MLP}} = 0.14$), confirming that low MLP- κ genuinely reflects low feature-subset quality rather than an estimator mismatch. Taken together, these results validate the use of a fixed, shared MLP as the uniform ranking criterion: the MLP-based ranking is not susceptible to inflation by methods that exploit estimator-specific noise.

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